01 / PRODUCTION PROPOSAL

Build a family, not a folder of random generations.



74 work packages preserve the original first-zone register. Only 20 have proposed Tripo candidate recipes; the rest cover exact modules, shared materials, graphics, bindings, animation, UI, audio or scene assembly.

Generate selectively. Construct exact parts. Validate in the engine.

This atlas and data/assets.json supersede v0.1 for production routing. It includes no finished meshes or audio. The templates authorize zero credits until the user approves a specific batch.

02 / ROUTE SELECTION

Use the right tool for the constraint.

Route	Packages	Responsibility
HYBRID TRIPO	12	Generate visual shell/candidate; exact controls, pivots and assembly follow in DCC.
STUDIO ASSEMBLY	3	Compose existing assets, not a new scene mesh.
DCC EXACT	24	Script/model exact reusable geometry; preserve grid, clearances and pivots.
TRIPO PROP	8	Candidate prop generation; inspect topology, cleanup, UV and texture delivery.
GAMEPLAY BINDING	6	Reuse assets; attach discovery, restoration or effect state.
MATERIAL GRAPHICS	8	Author shared trims, palettes, atlas artwork and readable graphics.
RIG ANIMATION	5	Purpose-built rig/animations with tested joints and state transitions.
UI CODE	4	Responsive UI and state feedback; no text baked by 3D generation.
AUTHORED AUDIO	4	Approved recorded/designed samples with provenance and matching previews.

An asset route is not permission to spend. P0 means core scope, not automatic batch execution. Use the already connected Tripo integration only after live capability and documentation checks. [T05, T06, O01]

03 / PROJECT BUDGETS

Measure the delivered model, not the advertised mode.

Asset/family	Full-asset triangle target	Texture delivery target	Important qualifier
H01 sampler	2500–4500	Shared 1024 + separate UI	Count every duplicated pad in the assembly.
H02 speakers	1000–2200	Shared 1024 equipment trim	Cone separate only where animation needs it.
H03 crate + insert	450–950	Shared wood + sleeve atlas	No mesh per record sleeve.
H04 robot	2500–5000	Shared 1024 + face atlas	Small rigid rig; no tread-link simulation.
H05 lamp	500–1100	Shared 512/1024	Visible insert and light attachment separate.
Architecture	Use per-ID cap; mostly 100–750	Shared 512 tile / 1024 trim	Exact joins and portals outrank generated detail.
Small/medium props	Use per-ID cap	Usually 256/512	Unique texture and MeshPart count matter too.

Single-mesh project cap: 10000 triangles without an explicit exception. Count after modifiers and triangulation; check imported mesh splits. These are authored constraints, not Roblox or Tripo platform maximums.

Default generation uses explicit face_limit and geometry-first output. Texture quality labels do not guarantee final pixel dimensions. [T01–T03, R02]

04 / HERO ASSET H01

MK-I sampler workstation



Reference crop: style authority only, not a ready-to-submit isolated image.

Generate the housing only.

Build 16 pads, screen, labels, knob interfaces and targets deterministically. Maintain canonical 01–16 order. Keep a clean control deck so rebuilding is straightforward.

Route: HYBRID TRIPO
Bounds: 6 x 1.25 x 4 studs
Full-asset budget: 2500–4500 triangles
Candidate face cap: 2200

Construction contract

Pivot: Bottom-center; controls face -Z
Collision: One optional query proxy; no decorative collisions
Separated/moving: 16 logical pad targets; screen plane; optional pad travel

State and acceptance

Dormant / ready / recording / playing / unavailable. 16 correctly mapped pads; screen separate; readable at play distance; no baked labels needed for gameplay.

Dependencies: T01; G03; U01

Generate or model only the housing; create exact 4 x 4 controls deterministically. Full assembled asset cap 4500 triangles, including repeated pad instances.

05 / HERO ASSET H02

Compact speaker stack



Reference crop: style authority only, not a ready-to-submit isolated image.

Preserve the woofer-and-horn silhouette.

A simple cabinet can be shared. Separate a cone only if needed for restrained local animation. Do not make the collision vibrate or use modeled grille holes everywhere.

Route: HYBRID TRIPO
Bounds: 4 x 8 x 3 studs
Full-asset budget: 1000–2200 triangles
Candidate face cap: 1800

Construction contract

Pivot: Floor-center; cone forward -Z
Collision: Simple box, not cone mesh
Separated/moving: Woofer / horn face targets; no physics vibration

State and acceptance

Quiet / active; one cabinet reused. Stacking socket aligned; cone child named; no animated collision; quiet state visually complete.

Dependencies: T01

Runtime measurements and Studio approval are required. Existing dimensions and caps are proposed project targets.

06 / HERO ASSET H03

Vinyl crate and record insert



Reference crop: style authority only, not a ready-to-submit isolated image.

Empty crate plus one grouped insert.

Generate/model the shell; author an insert using original sleeves and the shared G01 atlas. A browse proxy stays independent of visible records.

Route: HYBRID TRIPO
Bounds: 3 x 2.5 x 3 studs
Full-asset budget: 450–950 triangles
Candidate face cap: 650

Construction contract

Pivot: Floor-center
Collision: Box for solid placed crates; inserts noncolliding
Separated/moving: Separate record insert only

State and acceptance

Empty / filled; sleeves share atlas. Empty shell plus one insert delivered; artwork original; browse proxy independent of insert.

Dependencies: T02; G01

Runtime measurements and Studio approval are required. Existing dimensions and caps are proposed project targets.

07 / HERO ASSET H04

Maintenance helper robot



Reference crop: style authority only, not a ready-to-submit isolated image.

A candidate is not a rig.

Use a neutral pose with arms clear of the body. Rebuild/split rigid parts and author head/arm pivots. Keep the face panel replaceable. Avoid automatic humanoid assumptions.

Route: HYBRID TRIPO
Bounds: 3.5 x 4 x 3 studs
Full-asset budget: 2500–5000 triangles
Candidate face cap: 3500

Construction contract

Pivot: Root at floor-center; face -Z
Collision: Single stable body proxy; cosmetic wheels
Separated/moving: Root, head yaw/pitch, two arm pivots, wheel/tread surfaces

State and acceptance

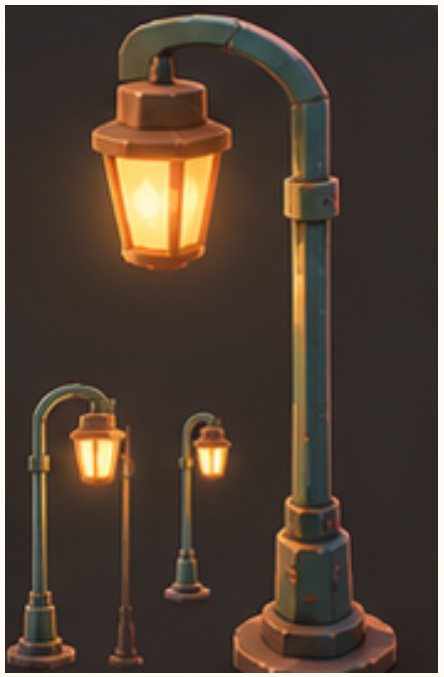
Idle / guide / listen / celebrate / waiting. Robot remains readable without glow; chirp interaction socket present; no per-link tread simulation.

Dependencies: T01; G03

Static Tripo output is a design source, not a rig. Split/retopologize rigid parts, author pivots and rig, then count full asset triangles.

08 / HERO ASSET H05

Curved street lamp



Reference crop: style authority only, not a ready-to-submit isolated image.

One coherent lamp family.

Use a sturdy curved pole, broad base and readable lantern. Create separate bulb/insert and light attachment; it must still read with local lighting and bloom disabled.

Route: HYBRID TRIPO
Bounds: 2.5 x 12 x 3 studs
Full-asset budget: 500–1100 triangles
Candidate face cap: 900

Construction contract

Pivot: Floor-center
Collision: Pole/base box only
Separated/moving: Bulb/glass target; light attachment

State and acceptance

Dormant / restored; light states not new meshes. Emissive insert separate from glass; lit silhouette readable with local light disabled.

Dependencies: T01

Runtime measurements and Studio approval are required. Existing dimensions and caps are proposed project targets.

09 / HERO ASSET H06

Record-shop / workshop shell



Reference crop: style authority only, not a ready-to-submit isolated image.

Assemble the shop from exact modules.

Never generate the building, pavement, props and plants as a fused diorama. Preserve playable interior, portals, ramp and camera clearance. Restoration reuses the shell.

Route: STUDIO ASSEMBLY
Bounds: 32 x 16 x 24 studs
Full-asset budget: Assembly: sum the visible components
Candidate face cap: No Tripo job

Construction contract

Pivot: Front-left floor grid corner
Collision: Modular proxies; portal remains clear
Separated/moving: Door and optional shutter; sign face separate

State and acceptance

Dormant / restored dressing; shared shell. One modular shell, not a fused diorama; 24 x 20 usable interior; doorway and camera clearances pass.

Dependencies: M01; M03; M04; M05; M06; M07; M08; M10; M11; M12; M17; T02; T03; T04

Build from exact kit modules. Never generate a fused storefront scene or its playable collision.

10 / EXACT MODULAR KIT

Structure set A: buildability first.

ID	Module	Proposed bounds / studs	Tri cap	Key construction detail
M01	Full wall bay	8 x 12 x 1	350	Teal broad panel with restrained seams; no tiny brick geometry.
M02	Half wall bay	4 x 12 x 1	250	Half-width infill matching M01 edges.
M03	Corner post	1 x 12 x 1	220	Cream/wood corner reinforcement; clean 90-degree fit.
M04	Door portal bay	8 x 12 x 1	450	Clear opening 6 wide x 10 high; no threshold trip lip.
M05	Display window bay	8 x 12 x 1	550	Large display opening; interior dark fill permitted for distant duplicate.
M06	Door leaf	6 x 10 x 0.5	400	Hinge
M07	Floor tile	8 x 1 x 8	120	Flush edges and simple collider; surface detail through material.
M08	Roof panel	8 x 1 x 8	160	Simple top plane with visible edge only; no inaccessible interior geometry.
M09	Roof-edge trim	8 x 1 x 1	220	Broad metal coping; matches shared trim scale.

All modules use DCC_EXACT. Lower-left corner pivots and a shared grid prevent cumulative drift. Validate adjacent bays, a 90-degree corner, a door leaf and a window before extending the kit.

Do not use a generated wall’s approximate dimensions as the grid definition.

11 / EXACT MODULAR KIT

Structure set B: circulation and identity.

ID	Module	Proposed bounds / studs	Tri cap	Key construction detail
M10	Striped awning bay	8 x 2 x 4	650	Soft scalloped silhouette and broad stripe rhythm.
M11	Entry step unit	8 x 2 x 4	220	Chunky warm stone; pair with collision ramp.
M12	Access ramp	8 x 2 x 8	100	Same paving as steps; visually intentional, not an invisible workaround.
M13	Paving tile	8 x 0.5 x 8	120	Broad stone blocks; minimal geometry cracks; no ankle-height edges.
M14	Curb section	8 x 1 x 1	160	Slightly softened sandstone edge; shallow dirt at base only.
M15	Waterfront railing	8 x 4 x 0.5	450	Simple chunky posts; backstop geometry separate from visible bars.
M16	Roof parapet	8 x 3 x 1	250	Low warm masonry border; blocked roof is not promised as a playable area.
M17	Vinyl storefront sign	8 x 8 x 1	750	Black record disc with amber label; broad grooves in texture, not geometry.
M18	Shop shutter	8 x 10 x 0.5	300	Shutter transform

M10 awning, M17 vinyl sign and warm openings carry the shop identity. Keep paths and mounting interfaces exact. M18 shutter is optional P2; do not delay the playable workshop for it.

Visible depth should come from strong form and light, not a high count of tiny architectural meshes.

12 / SUPPORTING PROPS

Workshop furniture and everyday objects.

ID	Prop	Route	Tri cap	Production instruction
P01	Studio workbench	TRIPO PROP	900	H01 footprint fits; standing interaction needs no seat.
P02	Round workshop stool	TRIPO PROP	450	Optional seating only; cannot block station.
P03	Listening bench	TRIPO PROP	650	Sightline to stage; accessible approach.
P04	Record browsing bin	HYBRID TRIPO	800	Browse socket on front; artwork legible at close range.
P05	Wall shelf	TRIPO PROP	300	Mounted off head/camera route; shared props not embedded duplicates.
P06	Storage crate	TRIPO PROP	350	Box collider; repeat uses same mesh ID.
P07	Headphones	TRIPO PROP	1000	Reads as headphones at medium shot; does not become a functional mic promise.
P08	Woven studio rug	DCC EXACT	24	No floor z-fighting; noncolliding.

Generate only selected candidates after the hero family passes. Simple geometry may be built deterministically when it is cheaper or more exact. Reuse T01–T04 materials; keep micro-props out of the first functional milestone.

13 / SUPPORTING PROPS

Utilities, plants and supporting detail.

ID	Prop	Route	Tri cap	Production instruction
P09	Cable-run kit	DCC EXACT	220	No physics cables; simplified at distance; route does not imply a puzzle unless tagged.
P10	Cable coil	DCC EXACT	450	Noncolliding; removable without losing scene meaning.
P11	Planter pot	TRIPO PROP	400	Foliage socket centered; separate plant reusable.
P12	Broad-leaf plant cluster	TRIPO PROP	550	Low transparency layering; no leaf collisions.
P13	Wall / pendant lamp family	HYBRID TRIPO	500	Shared lamp core; light attachment consistent.
P14	Utility junction box	HYBRID TRIPO	400	Front socket and status insert separate.
P15	Repair-bench clutter cluster	DCC EXACT	650	Single dressing cluster; does not block sampler or become a loot promise.
P16	Rooftop vent / duct	HYBRID TRIPO	500	Discovery version reuses master and adds I02 binding.

Generate only selected candidates after the hero family passes. Simple geometry may be built deterministically when it is cheaper or more exact. Reuse T01–T04 materials; keep micro-props out of the first functional milestone.

14 / INTERACTIVE AND ENVIRONMENTAL WORK

Reuse a model before commissioning another.

ID	Package	Type / route	Acceptance focus
I01	Metal strike-sign discovery node	HYBRID TRIPO	One-shot sample ID binding; preview and collect states; no need to hit a tiny target.
I02	Vent-thump discovery binding	GAMEPLAY BINDING	No new vent mesh; audio cue, label and collection badge agree.
I03	Robot-chirp discovery binding	GAMEPLAY BINDING	Approved authored chirp sample linked; collected state not repeated as reward.
I04	Workshop resonator junction	HYBRID TRIPO	All four visible states supplied; restoration does not permanently alter public hub for others.
I05	Listening-stage control podium	HYBRID TRIPO	Join/listen action readable; mute/leave available without needing podium again.
W01	Courtyard listening-stage assembly	STUDIO ASSEMBLY	v0.2 overrides the oversized v0.1 stage proposal: prototype a 12 x 8 stage within a 28 x 24 courtyard; verify audience and navigation clearance before art lock.
W02	Waterfront edge and calm water	STUDIO ASSEMBLY	No unmarked fall route; no expensive reflective requirement for readability.
W03	Distant skyline cluster	DCC EXACT	Horizon adds depth without stealing contrast from shop or implying accessible doors.
W04	Elevated transit / bridge silhouette	DCC EXACT	Readable only as backdrop; removable on low tier.

I02 reuses P16. I03 reuses H04. Bindings are not new meshes, and no image-to-3D job supplies the discovery audio.

15 / SHARED APPEARANCE

Materials and graphics are production systems.

ID	Package	Purpose	Acceptance
T01	Equipment paint / trim material set	Teal, cream, graphite, small coral accents; authored roughness/metalness; reusable equipment trim.	Check neutral light and dusk; maps named; no unnecessary maps or duplicate uploads.
T02	Wood material set	Broad warm grain; no baked sunlight; shared crates, benches, shelves.	Check neutral light and dusk; maps named; no unnecessary maps or duplicate uploads.
T03	Masonry / paving material set	Warm sandstone and desaturated plaster; restrained joints and edge dirt.	Check neutral light and dusk; maps named; no unnecessary maps or duplicate uploads.
T04	Fabric / rubber material set	Coral/cream awning stripe, matte pad rubber, subdued rug weave.	Check neutral light and dusk; maps named; no unnecessary maps or duplicate uploads.
G01	Original record sleeve / label atlas	Abstract circles, stripes, playful local motifs; no real album covers or artist logos.	Editable vector/raster source; transparent padding; no baked essential language.
G02	Neighborhood poster / sticker atlas	Repair notices, abstract music graphics, crate marks; optional flavor text only.	Editable vector/raster source; transparent padding; no baked essential language.
G03	Equipment face / pad / icon atlas	Warm bar eyes and clear transport symbols; shape accompanies color.	Editable vector/raster source; transparent padding; no baked essential language.
G04	Wayfinding and status graphics	Consistent rounded icon family; actual names/status text in localizable UI.	Editable vector/raster source; transparent padding; no baked essential language.

Tripo output does not automatically use these shared atlases. Re-UV/rebake explicitly, and verify map channels in Studio. Keep gameplay labels and localized text out of generated textures. [R01, R02]

16 / MOTION AND EFFECTS

Small reactions. Explicit state ownership.

ID	Package	Dependencies	Acceptance
A01	Sampler pad feedback	H01; U01	Loops seamless where needed; stop/rejoin safe; reduced-motion fallback supplied.
A02	Speaker cone response	H02	Loops seamless where needed; stop/rejoin safe; reduced-motion fallback supplied.
A03	Robot basic motion set	H04	Loops seamless where needed; stop/rejoin safe; reduced-motion fallback supplied.
A04	Robot celebration / carry set	H04; A03	Loops seamless where needed; stop/rejoin safe; reduced-motion fallback supplied.
A05	Restoration reveal choreography	H06; I04; F01	Loops seamless where needed; stop/rejoin safe; reduced-motion fallback supplied.
F01	Dusk and practical-light presets	H05; P13	Scene remains complete when disabled; no gameplay information conveyed solely by effect.
F02	Music-reactive visual preset	H02; H05; I04	Scene remains complete when disabled; no gameplay information conveyed solely by effect.
F03	Discovery feedback	I01; I02; I03; U02	Scene remains complete when disabled; no gameplay information conveyed solely by effect.
F04	Ambient dust / water accents	F01; W02	Scene remains complete when disabled; no gameplay information conveyed solely by effect.

Face, head and cone motions are local visual reactions. They must never start the next beat or become authoritative musical state. Effects-disabled and low-quality presentation remain complete and usable.

17 / USER INTERFACE AND AUDIO

The instrument needs more than a convincing shell.

ID	Package	Route	Acceptance
U01	Workstation UI art kit	UI CODE	Touch and keyboard focus states; text labels localizable; monochrome/reduced-motion readability.
U02	Discovery / sample card kit	UI CODE	Touch and keyboard focus states; text labels localizable; monochrome/reduced-motion readability.
U03	Listening / session status kit	UI CODE	Touch and keyboard focus states; text labels localizable; monochrome/reduced-motion readability.
U04	Save / restore feedback kit	UI CODE	Touch and keyboard focus states; text labels localizable; monochrome/reduced-motion readability.
S01	Starter production sound bank	AUTHORED AUDIO	Source/rights manifest, loudness review, trims, import permissions and published-place playback checked.
S02	Discovery sample bank	AUTHORED AUDIO	Source/rights manifest, loudness review, trims, import permissions and published-place playback checked.
S03	Interaction and UI cue bank	AUTHORED AUDIO	Source/rights manifest, loudness review, trims, import permissions and published-place playback checked.
S04	Quiet waterfront ambience	AUTHORED AUDIO	Source/rights manifest, loudness review, trims, import permissions and published-place playback checked.

Use original/authorized sound assets and record provenance. Discovery preview and collected sample must match. Exact UI layout, transport, save and session behavior remain code-driven. This art pack does not contain final audio or implement the music engine.

Do not treat a generated picture of a display as a working display.

18 / PROVIDER POLICY

Game optimization must be explicit.

Route/profile	Initial policy	Must not happen
P1 geometry-first	Pin the verified P1 snapshot; explicit per-asset face_limit; texture=false and pbr=false. [T01,T02]	Sending H-series smart_low_poly, parts, quad or ultra-geometry fields to P1.
H-series fallback	Use compatible smart_low_poly=true, explicit cap, reviewed endpoint and revised cost. [T03]	Silently falling back to a dense default or combining low-poly and segmented-parts modes.
P2 Preview	Disabled by default; opt-in only after complete current schema and cost review. [T04]	Selecting the newest Preview just because its maximum face count is higher.
CLI game-mobile	Inspect resolved preset/options and enforce our tighter asset caps. [T05]	Assuming a preset name proves Roblox readiness or inventing an undocumented dry-run flag.

Research snapshot: 22 October 2026. P2/retopology documentation revisions disagree in retrieved material. The runtime must refresh the official pages and actual connected schema, then record a receipt. This pack has not authenticated to the user's Tripo account.

If the connected tool cannot enforce the budget, do not call its unbounded default.

19 / DCC AND ENGINE ACCEPTANCE

Six gates separate a candidate from a shipped asset.

Gate	Evidence required	Failure action
G0 / Preflight	Fresh docs and schema; approved reference; explicit fields; account, region and spend cap	No paid call; resolve only the blocked operation.
G1 / Candidate	Downloaded real files; real task ID; input/output hashes; credits and lineage	Keep failed candidate/reason; do not silently regenerate.
G2 / DCC clean	Measured dimensions/triangles/maps; exact pivots, parts, UVs and collision	Repair deterministic issues before adding provider jobs.
G3 / Art approved	Neutral, dusk, gameplay-distance and effects-off comparisons	Fix silhouette/material mismatch, not only beauty lighting.
G4 / Studio imported	Actual IDs, ownership, hierarchy, scale, materials, proxies and warnings	Do not mark in-game ready from a DCC render.
G5 / Runtime verified	Recorded device/quality scene profile; two-player states and usable controls	Report missing evidence; no invented FPS or completion.

Initial scene targets are diagnostic only: 150000 visible environment triangles, 250 visible environment MeshParts and two shadow-casting local lights affecting the main view. Measure actual cost with gameplay, UI and audio running. [R03]

20 / PROVENANCE AND HANDOFF

A documented starting point, not a frozen API contract.

ID	Primary source	Official source link
T01	Tripo P1 model card	developers.tripo3d.ai/en/models/p1
T02	Tripo P-series image generation	developers.tripo3d.ai/en/docs/generation-image-to-model/p
T03	Tripo H-series generation	developers.tripo3d.ai/en/docs/generation-text-to-model/standard
T04	Tripo P2 model card	developers.tripo3d.ai/en/models/p2
T05	Tripo CLI	developers.tripo3d.ai/en/docs/cli
T06	Tripo Codex plugin	developers.tripo3d.ai/en/docs/codex-plugin
T07	Tripo task query	developers.tripo3d.ai/en/docs/task-query
T08	Tripo billing	developers.tripo3d.ai/en/docs/billing
T09	Tripo retopology	developers.tripo3d.ai/en/docs/mesh-decimate
T10	Tripo conversion	developers.tripo3d.ai/en/docs/models-convert
R01	Roblox Importer	create.roblox.com/docs/studio/importer
R02	Roblox texture specifications	create.roblox.com/docs/art/modeling/texture-specifications
R03	Roblox performance guidance	create.roblox.com/docs/performance-optimization/improve
O01	Codex MCP configuration	developers.openai.com/codex/mcp

Full source notes, fetch limitations and known conflicts are in docs/SOURCES_AND_DOCS_DRIFT.md. T11–T14 cover multiview, texturing, the older MCP bridge and the current JS SDK; O02 covers OpenAI documentation access. Recheck current docs before each production batch.

Reference board: supplied and selected by the user. Specifications: authored project proposals. No new 3D assets, audio, provider charges, authenticated connection checks, Studio imports or target-device benchmarks are claimed by this handoff.